

Volatility Forecasting: GARCH and Recurrent Models

A controlled walk-forward comparison
with calibration and uncertainty diagnostics

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EXPERIMENT SCOPE

Nine specifications, 3,738 common test dates, February 5, 2010 to December 11, 2024. Six neural models each trained across 178 expanding-window folds. One primary training seed.

Research question

Can recurrent models improve one-day equity-index variance forecasts over parsimonious classical baselines? The comparison separates implementation correctness, forecast positivity, mean calibration, and loss-based performance.

What changed

The rerun repairs daily GARCH state updates, corrects a target inverse mismatch, aligns neural training availability, seeds before model construction, and evaluates train/validation loss at the same restored checkpoint. Positive log-ratio hybrids replace the original unconstrained additive residual specification.

Main finding

GARCH-t has the lowest QLIKE point estimate (-8.4613); GARCH-t has the lowest MSE (1.601e-07). The best neural QLIKE point estimate is Ratio LSTM (-6.6595), higher than GARCH-t (-8.4613).

Across the nine complete prediction series, 0 forecasts are at or below $1e-12$. Positivity does not establish calibration or superiority. The study uses a previously examined historical snapshot, not an untouched prospective holdout.

Source: corrected-v1 aggregate results; cached ^GSPC snapshot; no fresh market-data download.

01 / A Comparable Experiment

Element	Corrected design
Data and target	Cached daily ^GSPC prices, 2003-2024. Target y is squared log return in daily decimal-return variance units.
Classical baselines	Student-t GARCH(1,1); lagged mean of 21 squared returns; EWMA with lambda 0.94.
Neural inputs	21 rows ending at t-1: log return, squared return, absolute return, annualized 21-day rolling volatility. Feature/ratio hybrids also include lagged GARCH variance.
Common availability	Modeling frame begins after the 756-return GARCH warmup. Initial neural training span has 777 rows, yielding 756 sequences; validation has 252 observations.
Walk-forward tests	178 non-overlapping blocks of 21 observations. Training expands; validation and model weights are reselected per fold. The incomplete trailing block is excluded.
Neural settings	8 units; dropout 0.1; Adam 0.001; batch 64; maximum 35 epochs; patience 6; seed 42 plus split ID. CPU execution; fixed-length recurrence unrolled.

Forecast timing

A forecast for date t is issued using information through t-1. Feature and target transform statistics are fitted only on the training sample. The validation sample selects early stopping. Later folds may train on observations that were test data in an earlier fold, once those observations become historical.

Why the evaluation now begins in 2010

The original hybrid runs had very little early training history after GARCH warmup. The rerun requires a full shared neural training span and therefore starts testing later. Old and new loss levels must not be interpreted as a one-change before/after comparison. GARCH retains its own expanding return-estimation history; equal neural dates do not make every estimator's effective memory identical.

Squared returns are a noisy second-moment proxy. Their interpretation as conditional variance additionally assumes a zero conditional mean. The 2008 crisis is in training history, not this corrected test period.

02 / Results on Identical Test Dates

All rows use the same 3,738 targets and dates. Three classical baselines and six freshly trained neural specifications are included. This is a new historical walk-forward run, not a rescore of the old forecasts.

Forecast	MSE / 10 ⁻⁷	QLIKE	Mean h / mean y
GARCH-t	1.60062	-8.46125	1.038
Lagged RV21	1.87208	-8.35289	1.005
EWMA 0.94	1.74818	-8.41999	1.004
Pure LSTM	1.93876	-6.44830	0.258
Pure GRU	1.91925	-6.43031	0.259
Feature LSTM	1.96415	-6.60657	0.262
Feature GRU	1.97011	-6.51491	0.256
Ratio LSTM	1.86316	-6.65949	0.299
Ratio GRU	1.85128	-6.57178	0.283

$$\text{MSE} = \text{mean}[(y - h)^2]$$

$$\text{QLIKE} = \text{mean}[\log(h) + y/h]$$

Both losses are lower-is-better. QLIKE uses a fixed 1e-12 numerical floor for y and h; no floor was selected against test losses. Negative QLIKE levels are normal under this convention. The MSE column is scaled by 10⁻⁷; these are not returns or percentages.

Interpretation

GARCH-t has the lowest QLIKE point estimate (-8.4613); GARCH-t has the lowest MSE (1.601e-07). The best neural QLIKE point estimate is Ratio LSTM (-6.6595), higher than GARCH-t (-8.4613).

Mean h / mean y is a coarse level-calibration diagnostic, not a full conditional calibration test. A value below one means the average variance prediction is below the average realized proxy. A good average ratio does not rule out tail or regime failures.

Both original-scale MSE and QLIKE can be robust to an unbiased noisy proxy under suitable conditions [2]. Different loss weighting and finite-sample behavior can produce different rankings. Point-estimate selection across several candidates does not establish statistically reliable superiority.

03 / Paired Loss Uncertainty

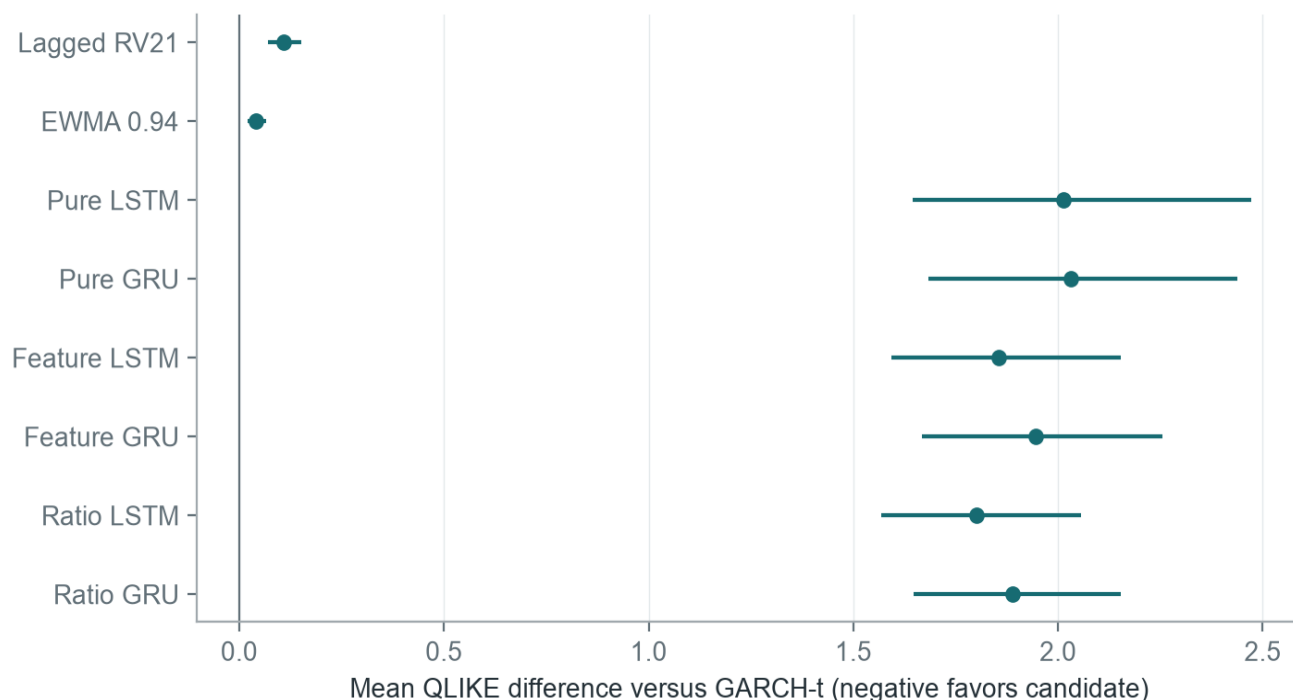


Figure 1. Candidate minus GARCH-t mean QLIKE, with 95% pointwise percentile intervals from a paired stationary bootstrap. Average block length 21; 2,000 replications; seed 20260924. Negative values favor the candidate. All comparisons use 3,738 dates.

What the interval addresses

Daily candidate and benchmark losses are paired by date. Resampling blocks retains some serial dependence rather than treating daily observations as independent. The interval concerns the mean loss difference conditional on these trained forecasts and suitable time-series assumptions.

For Ratio LSTM, the observed QLIKE difference versus GARCH is **+1.8018**, with interval **[+1.5671, +2.0561]**. This model is highlighted because it has the lowest neural QLIKE point estimate; that selection is post hoc and is not accounted for by the individual interval.

What it does not address

These are not simultaneous intervals across all candidates and do not quantify neural initialization uncertainty, data-vendor variation, research selection, or future regime shifts. Block-length sensitivity at 5 and 63 is retained in the aggregate evidence file. A single historical comparison is not proof of deployment value.

04 / Positivity Is Not Calibration

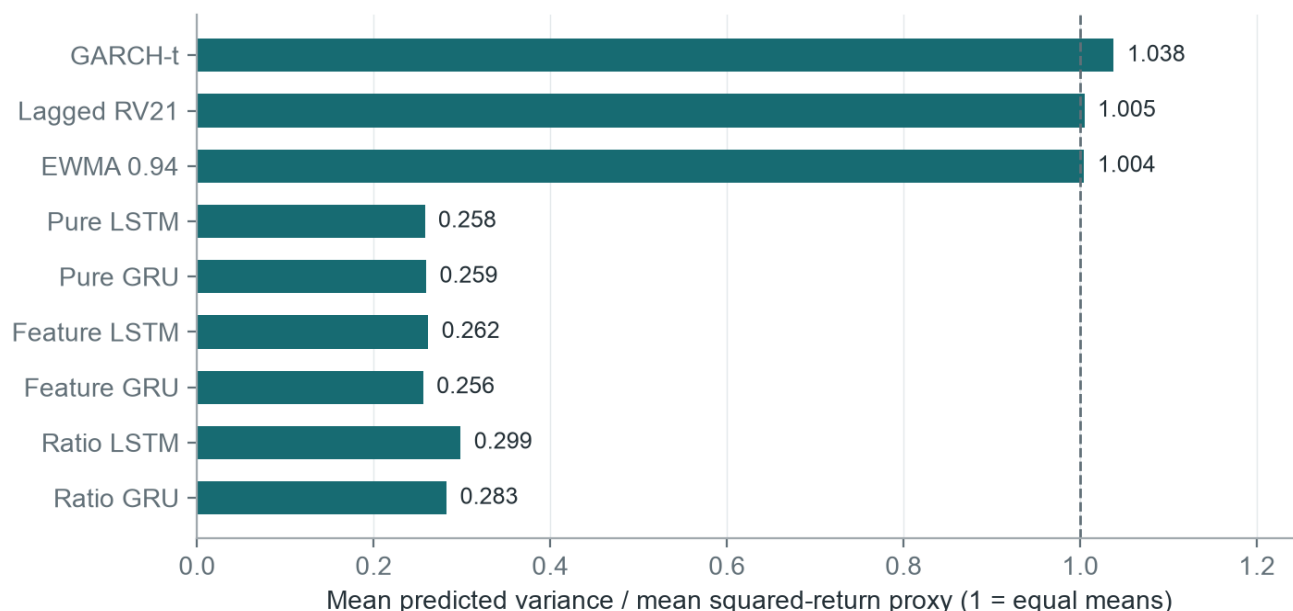


Figure 2. Mean predicted variance divided by mean squared-return proxy on the common test sample. The dashed line marks equal means; it is not a significance threshold.

A positive hybrid construction

The ratio hybrid learns $z = \log(\max(y, 1e-8)/g)$, where g is GARCH variance, and reconstructs $h = g \exp(\text{predicted } z)$. Finite log predictions and positive g yield positive variance; numeric overflow or underflow is rejected. The target floor prevents $\log(0)$, and is distinct from an output floor.

The objective still matters

The pure/feature models train on standardized log targets; ratio models train on standardized log ratios. Squared error on $\log y$ targets $E[\log y \mid \text{information}]$. Exponentiating does not generally recover $E[y \mid \text{information}]$. Correcting the inverse transform therefore fixes a bug, not this estimand mismatch.

Under a conditional Gaussian illustration without clipping, $\exp(E[\log(r^2)])$ is approximately 0.281 times the conditional variance. This explains why well-fitted log-target models can underforecast arithmetic variance. It is not an empirical correction factor for this dataset.

A next controlled study should compare a positive output trained directly under QLIKE or another original-scale objective. No such additional model is claimed as completed here.

05 / Calendar-Period Diagnostics

The following calendar slices were fixed before inspecting the new test losses. They are descriptive partitions, not a fitted regime classification. Each column compares models on the same dates within that period.

Forecast	2010-2016	2017-2019	2020-2021	2022-2024
GARCH-t	-8.490	-8.950	-7.998	-8.211
Lagged RV21	-8.365	-8.816	-7.872	-8.182
EWMA 0.94	-8.458	-8.881	-7.906	-8.213
Pure LSTM	-6.887	-7.632	-4.054	-5.846
Pure GRU	-6.793	-7.664	-4.303	-5.772
Feature LSTM	-6.963	-7.570	-4.837	-5.995
Feature GRU	-6.842	-7.594	-4.894	-5.753
Ratio LSTM	-7.074	-7.307	-5.270	-5.973
Ratio GRU	-6.995	-7.284	-5.118	-5.844

Table 2. Mean QLIKE by period; lower is better. Observation counts: 2010-2016: 1,739; 2017-2019: 754; 2020-2021: 505; 2022-2024: 740.

Why inspect subperiods?

An aggregate mean can hide unstable relative performance or a handful of large loss contributions. Calendar diagnostics help identify questions for follow-up, but choosing favorable periods after seeing them would be a selection problem.

What remains untested

The main run uses one initialization schedule. It does not establish seed robustness, performance across assets, conditional calibration, or stability under a different target proxy. The historical dataset has been studied before; this is not an untouched final holdout.

Trading relevance

Variance forecasts may inform risk sizing, volatility targeting, or risk estimates. Those are potential applications, not results of this study. No strategy, execution model, transaction-cost analysis, portfolio P&L, or Sharpe-ratio improvement is claimed.

The appropriate conclusion is specific to the forecast target, information set, sample, objective, and baselines actually tested. More complex architecture is not itself evidence of better research or better investment performance.

06 / Recurrent Gates as Diagnostics

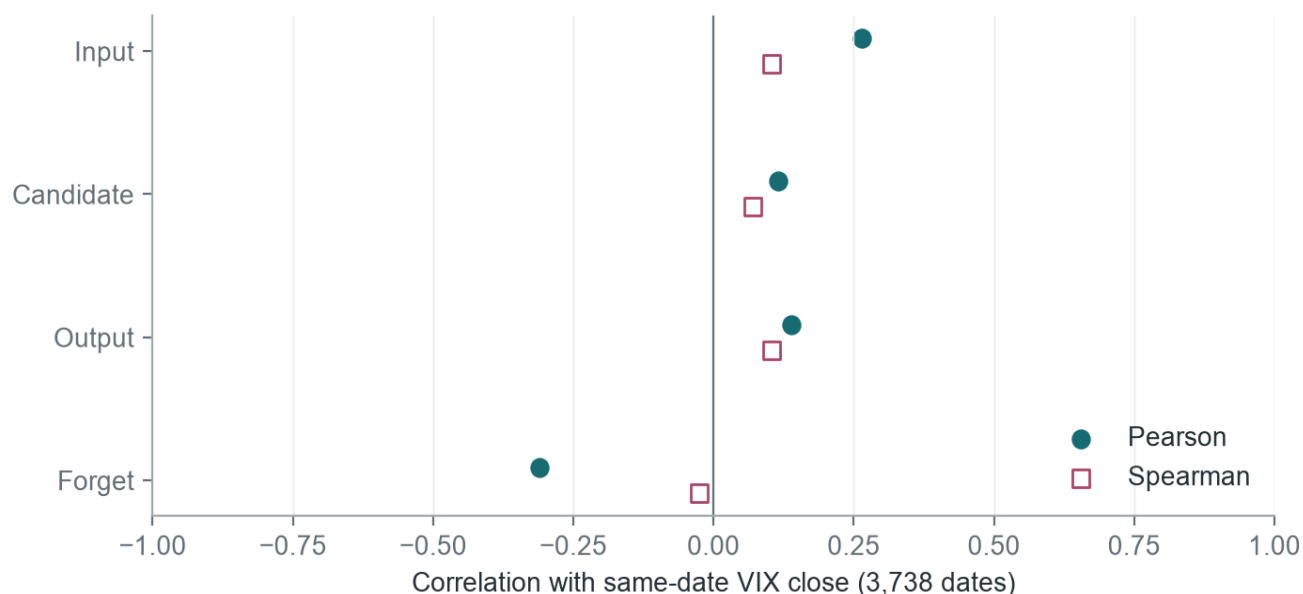


Figure 3. Corrected-run pure-LSTM activations, averaged across units and lags 1-20, versus same-date VIX close. Pearson and rank correlations use 3,738 matched dates. The appendix lists all 21 activation/variant summaries.

What is observable

The pure-LSTM forget-gate aggregate has Pearson correlation **-0.310** and Spearman correlation **-0.025** with VIX in this rerun. These are descriptive associations, not identified mechanisms.

Three interpretation limits

Timing: same-date VIX close is ex-post context and was not a predictor available at the previous-day forecast origin.

Pooling: the series combines separately trained models across folds; unit identities and fitted behavior need not remain stable.

Identification: an average activation does not identify a memory horizon, causal response, or tradable signal.

Stronger evidence would require controlled ablations, within-fold checks, multiple seeds, and correctly lagged predictive tests. Candidate activations are shown with the gate summaries for consistency; they are not sigmoid gates.

07 / Conclusions and Provenance

The corrected run demonstrates a reproducible research workflow with explicit forecast timing, shared neural sample availability, positive hybrid construction, simple baselines, and paired temporal uncertainty. The numerical and timing defects identified in the previous report have regression coverage.

GARCH-t has the lowest QLIKE point estimate (-8.4613); GARCH-t has the lowest MSE ($1.601e-07$). The best neural QLIKE point estimate is Ratio LSTM (-6.6595), higher than GARCH-t (-8.4613).

Next research priorities

First, align a positive variance output with an original-scale proper loss. Second, repeat across seeds and assets with a genuinely held-out evaluation plan. Third, connect forecasts to a prespecified economic objective only if the statistical evidence justifies the added complexity.

References

- [1] Bollerslev (1986), Generalized autoregressive conditional heteroskedasticity. [Journal of Econometrics](#).
- [2] Patton (2011), Volatility forecast comparison using imperfect volatility proxies. [Journal of Econometrics](#).
- [3] Hochreiter and Schmidhuber (1997), Long Short-Term Memory. [Neural Computation](#).
- [4] Cho et al. (2014), Learning Phrase Representations using RNN Encoder-Decoder. [arXiv:1406.1078](#).
- [5] [arch forecasting documentation](#) and [stationary bootstrap implementation](#).

Research record

The public repository includes code, tests, pinned research dependencies, this report builder, and aggregate results with input/code/forecast hashes. Raw market data and complete prediction CSVs remain local. The previous report is preserved in Git history; corrected-v1 is a separate experiment, not a silent overwrite of historical scores.

The underlying project originated in collaborative research. This revised portfolio report is presented by Eddie Liu. Contributor names and course identifiers are omitted; the byline does not imply sole authorship of the original experiments.

github.com/SRG-eddieliu/rnn-volatility-lab